

OLET1666 Assignment

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Abstract

This report presents analytical and numerical solutions to the one-dimensional diffusion equation governing carbon monoxide (CO) transport along a mine shaft of length $L = 2000$ m. An exact analytical solution is derived via separation of variables, decomposing the problem into a linear steady-state component and a transient Fourier series component. A Forward-Time Central-Space (FTCS) finite difference scheme is then implemented numerically and verified to be second-order accurate in space. The numerical scheme is applied to assess miner safety under a realistic evacuation scenario and subsequently extended to include a spatially varying source term modelling additional CO generation within the shaft.

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1 QUESTION 1: ANALYTICAL SOLUTION

1.1 Boundary and Initial Conditions

The boundary and initial conditions for this problem are as follows.

Boundary conditions:

- $x = 0$ (deep end): $C(0, t) = 1$ (fixed CO concentration)
- $x = L$ (surface): $C(L, t) = 0$ (fixed CO concentration)

Initial condition:

- $t = 0$: $C(x, 0) = 0$ (no CO initially present)

1.2 Solution Composition

The solution will be composed of two parts:

- A steady-state solution $C_S(x)$ representing the long-term CO distribution.
- A homogeneous solution $C_H(x, t)$ describing the transient behaviour towards the steady state.

1.3 C_S Solution

The diffusion equation:

$$\frac{\partial}{\partial t}C(x, t) = D \frac{\partial^2}{\partial x^2}C(x, t) \quad (1)$$

At a steady state, $\frac{\partial}{\partial t}C(x, t) = 0$, so equation (1) simplifies to:

$$0 = D \frac{d^2 C_S(x)}{dx^2} \quad (2)$$

Dividing through by D and integrating both sides:

$$A = \frac{dC_S(x)}{dx} \quad (3)$$

where A is a constant. Integrating once more:

$$C_S(x) = Ax + B \quad (4)$$

From the boundary conditions stated in Section 1.1:

$$C_S(0) = 1 \quad (5)$$

$$C_S(L) = 0 \quad (6)$$

Applying equation (5) to equation (4):

$$A(0) + B = 1 \implies B = 1 \quad (7)$$

Applying equation (6) to equation (4):

$$A(L) + 1 = 0 \implies A = -\frac{1}{L} \quad (8)$$

Substituting equations (7) and (8) into equation (4) gives the steady-state solution:

$$C_S(x) = 1 - \frac{x}{L} \quad (9)$$

Figure 1 displays a depiction of this solution.

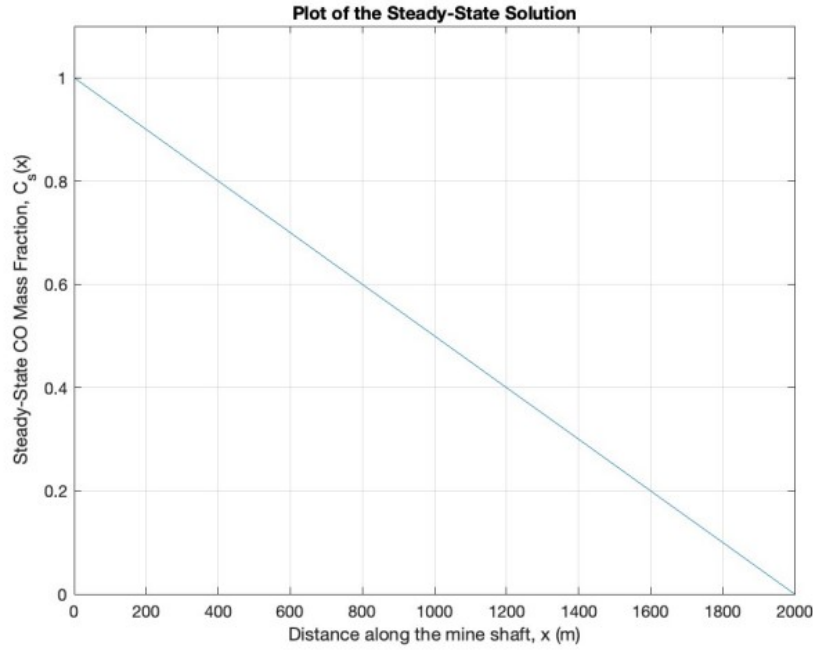


Figure 1: Sketch of the steady-state solution $C_S(x)$.

1.4 Initial Condition

Using superposition:

$$C(x, t) = C_H(x, t) + C_S(x) \quad (10)$$

Rearranging equation (10):

$$C_H(x, t) = C(x, t) - C_S(x) \quad (11)$$

Evaluating at $t = 0$ and applying the initial condition $C(x, 0) = 0$:

$$C_H(x, 0) = 0 - \left(1 - \frac{x}{L}\right) = \frac{x}{L} - 1 \quad (12)$$

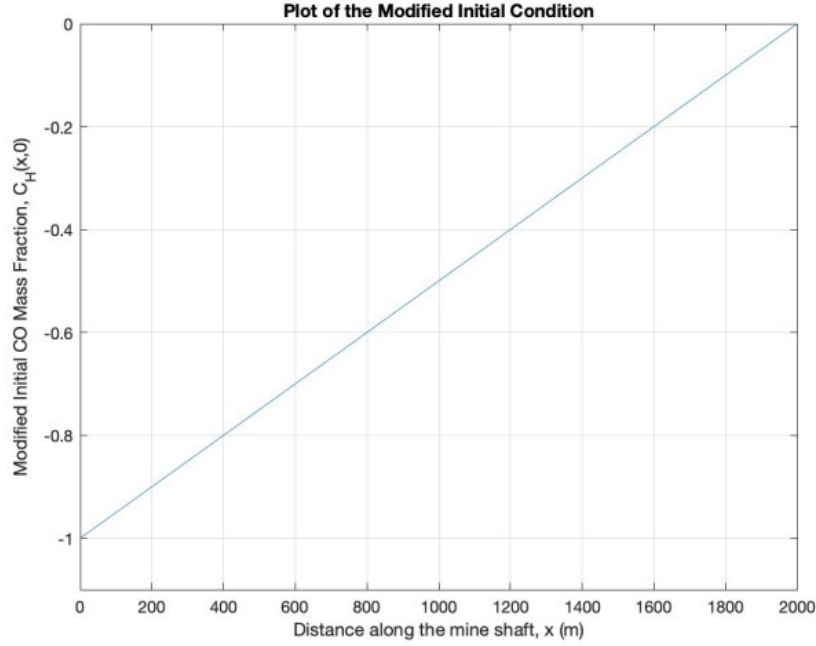


Figure 2: Sketch of the modified initial condition $C_H(x, 0)$.

1.5 Spatial and Temporal General Solutions

Applying separation of variables, $C_H(x, t) = F(x)G(t)$, yields two ODEs:

$$F''(x) + p^2 F(x) = 0 \quad (13)$$

$$G'(t) + Dp^2 G(t) = 0 \quad (14)$$

The general spatial solution to equation (13) is:

$$F(x) = A_p \cos(px) + B_p \sin(px) \quad (15)$$

Applying the homogeneous boundary condition $F(0) = 0$ to equation (15):

$$F(0) = A_p = 0 \quad (16)$$

so equation (15) reduces to $F(x) = B_p \sin(px)$. Applying $F(L) = 0$:

$$\sin(pL) = 0 \implies p = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots \quad (17)$$

giving the spatial eigenfunctions:

$$F_n(x) = B_n \sin\left(\frac{n\pi x}{L}\right) \quad (18)$$

The general temporal solution to equation (14) is:

$$G_n(t) = e^{-Dp^2 t} \quad (19)$$

Combining equations (18) and (19) and summing over all eigenmodes gives the general homogeneous solution:

$$C_H(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-D \frac{n^2 \pi^2}{L^2} t} \quad (20)$$

1.6 Fourier Coefficients

Evaluating equation (20) at $t = 0$ and matching with the modified initial condition from equation (12):

$$\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) = \frac{x}{L} - 1 \quad (21)$$

The Fourier sine coefficients are therefore:

$$B_n = \frac{2}{L} \int_0^L \left(\frac{x}{L} - 1\right) \sin\left(\frac{n\pi x}{L}\right) dx \quad (22)$$

Applying integration by parts with $u = \frac{x}{L} - 1$ and $dv = \sin\left(\frac{n\pi x}{L}\right) dx$:

$$du = \frac{1}{L} dx, \quad v = -\frac{L}{n\pi} \cos\left(\frac{n\pi x}{L}\right) \quad (23)$$

$$B_n = \frac{2}{L} \left(\left[-\frac{L}{n\pi} \cos\left(\frac{n\pi x}{L}\right) \left(\frac{x}{L} - 1\right) \right]_0^L + \frac{1}{n\pi} \int_0^L \cos\left(\frac{n\pi x}{L}\right) dx \right) \quad (24)$$

Evaluating the boundary term and noting $\sin(n\pi) = 0$:

$$B_n = \frac{2}{L} \left(0 - \left(-\frac{L}{n\pi} \right) + \frac{L}{n^2\pi^2} (\sin n\pi - \sin 0) \right) = \frac{2}{L} \cdot \left(-\frac{L}{n\pi} \right) = -\frac{2}{n\pi} \quad (25)$$

Now that we have B_n , the complete homogeneous solution from equation (20) becomes:

$$C_H(x, t) = \sum_{n=1}^{\infty} -\frac{2}{n\pi} \sin\left(\frac{n\pi x}{L}\right) e^{-D \frac{n^2\pi^2}{L^2} t} \quad (26)$$

1.7 MATLAB

As the number of eigenfunctions summed in the Fourier Series rises, the accuracy of the approximation of the modified initial condition also increases. In Figure 3, we see that when the number of Fourier terms is 1 the accuracy is very low, and it steadily rises when increased to ten terms, becoming highly accurate at 100 terms with the approximation converging almost immediately on the exact value. The code for this implementation can be located in the Appendix under 1.7.

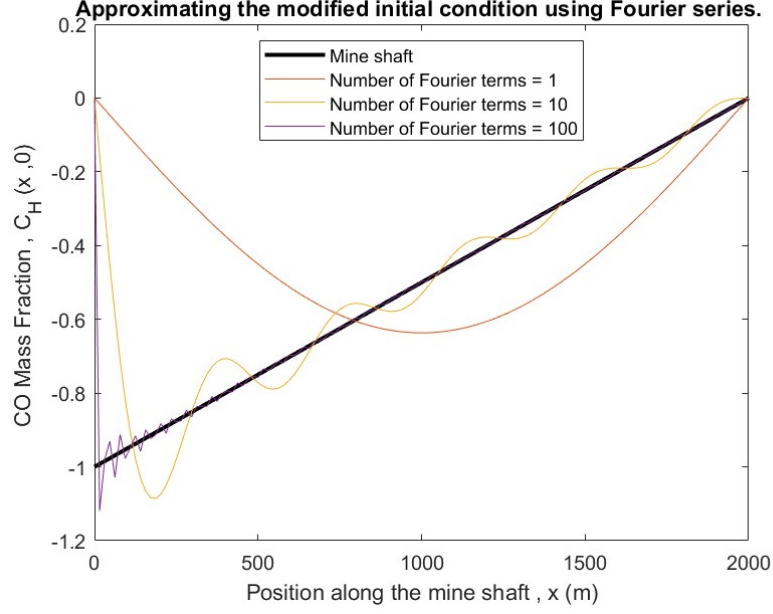


Figure 3: Approximating the modified initial condition using Fourier series.

1.8 Final Solution

The final solution $C(x, t)$ is given by combining equations (9) and (26) via equation (10):

$$C(x, t) = 1 - \frac{x}{L} + \sum_{n=1}^{\infty} -\frac{2}{n\pi} \sin\left(\frac{n\pi x}{L}\right) e^{-D \frac{n^2 \pi^2}{L^2} t} \quad (27)$$

Substituting $L = 2000$ m, the final solution is:

$$C(x, t) = 1 - \frac{x}{2000} + \sum_{n=1}^{\infty} -\frac{2}{n\pi} \sin\left(\frac{n\pi x}{2000}\right) e^{-D \frac{n^2 \pi^2}{2000^2} t} \quad (28)$$

2 QUESTION 2: NUMERICAL SOLUTION

2.1 Forward-Time Central-Space (FTCS) scheme

A Forward-Time Central-Space (FTCS) scheme is obtained by discretizing equation (1) using finite difference approximations. The forward finite difference approximation for the time derivative is:

$$C_t|_i^n \approx \frac{C_i^{n+1} - C_i^n}{\Delta t} \quad (29)$$

The central difference approximations for the spatial derivative are:

$$C_x|_{i-\frac{1}{2}}^n \approx \frac{C_i^n - C_{i-1}^n}{\Delta x}, \quad C_x|_{i+\frac{1}{2}}^n \approx \frac{C_{i+1}^n - C_i^n}{\Delta x} \quad (30)$$

Combining the two expressions in equation (30):

$$C_{xx}|_i^n \approx \frac{C_x|_{i+\frac{1}{2}}^n - C_x|_{i-\frac{1}{2}}^n}{\Delta x} = \frac{C_{i+1}^n - 2C_i^n + C_{i-1}^n}{(\Delta x)^2} \quad (31)$$

Substituting equations (29) and (31) into equation (1) and solving for C_i^{n+1} :

$$C_i^{n+1} = \sigma C_{i-1}^n + (1 - 2\sigma) C_i^n + \sigma C_{i+1}^n \quad (32)$$

where the diffusion number σ is defined as:

$$\sigma = \frac{D \Delta t}{(\Delta x)^2} \quad (33)$$

2.2 Explicit nature of the FTCS Scheme

The FTCS scheme given in equation (32) is referred to as *explicit* because it computes future values of C_i^{n+1} directly from the known values at the current time step n , without requiring the solution of a system of equations.

2.3 MATLAB Numerical Solution

Figure 4 shows the overlay of numerical solutions with varying values of n_x against the analytical solution from equation (28). The graph illustrates that the accuracy of the numerical solutions increases as n_x grows, producing closer alignment with the analytical answer.

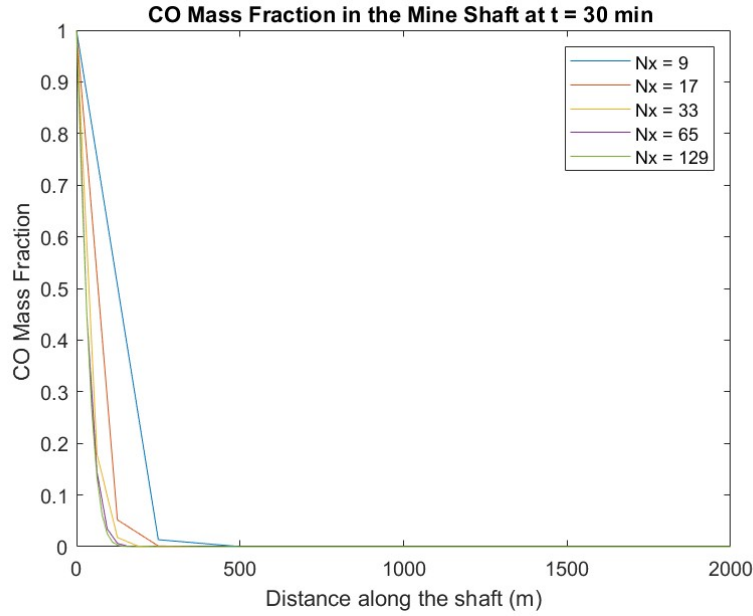


Figure 4: Comparison of the analytical solution with numerical solutions for various n_x values.

2.4 Verifying the Order of Accuracy of the numerical solution

The error norms and orders of accuracy are presented in Table 1. The MATLAB code responsible for generating these results may be found in the Appendix under 2.4. Because the FTCS scheme is second-order accurate in space, we expect the order of accuracy values to converge towards 2 as n_x increases. This is confirmed in Table 1.

Table 1: Error norms and orders of accuracy of the numerical solution (Appendix 2.4).

N	L^1	L^2	L^∞	$\mathcal{O}(L^1)$	$\mathcal{O}(L^2)$	$\mathcal{O}(L^\infty)$
9	3.018	2.012×10^{-1}	1.349×10^{-2}	—	—	—
17	5.891	5.285×10^{-1}	4.871×10^{-2}	-1.052	-1.519	-2.018
33	3.304	3.223×10^{-1}	3.875×10^{-2}	0.872	0.745	0.345
65	7.750×10^{-1}	7.409×10^{-2}	9.926×10^{-3}	2.139	2.169	2.009

2.5 CO mass fraction along the mine shaft

Figure 5 below contains a plot showing the numerical solution at times 1, 6, 12, 18, and 24 hours. This plot shows how as time increases, the portion of the mine shaft with a higher concentration of CO increases in length. However, the CO mass fraction decreases from 1 at $x = 0$ to 0 at various lengths of x for each time period.

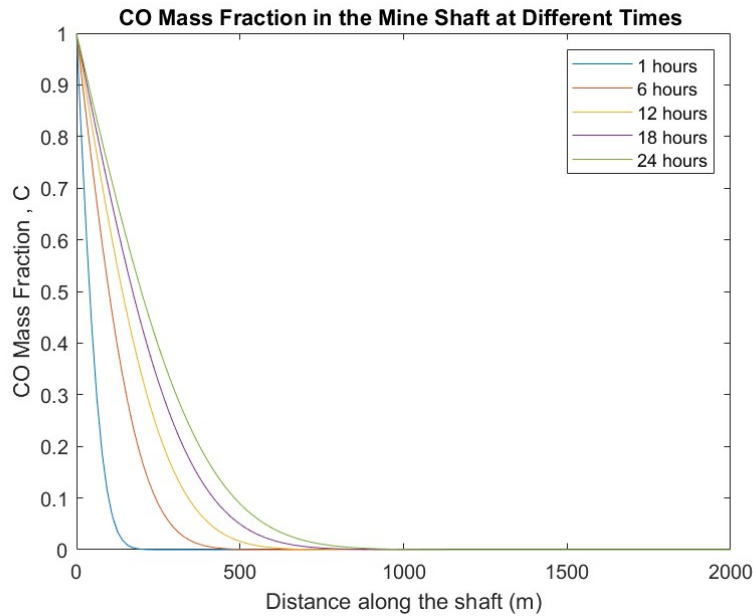


Figure 5: Numerical solution showing CO mass fraction along the mine shaft at different times.

Additionally, the gradient decreases with longer time periods, signifying more gradual concentration gradients than at shorter time periods. This is because the diffusion process results in the spread of CO from areas of high concentration to areas of low concentration, so the concentration gradient flattens out over time as the CO makes its way up the mine shaft.

2.6 Miners reaching the surface safely

The miners start at $x = 750$ m and an alarm is located at $x = 1000$ m. The alarm sounds when the CO mass fraction $C(x, t)$ reaches 0.001. The miners at 750 m evacuate at a speed of 360 m/hr upon hearing the alarm. A miner will die if the CO mass fraction exceeds 0.02, or if the CO mass fraction exceeds 0.001 for more than 12 hr.

Using the MATLAB code in the Appendix (2.6), the time at which $C(750, t) = 0.001$ is found to be 14.42 hr (equivalent to 51 900 s). The alarm sounds when $C(1000, t) = 0.001$, which occurs at 25.63 hr (92 280 s). Therefore, when the alarm sounds, the miners have already been exposed to a CO concentration exceeding 0.001 for 11.21 hr (40 380 s). After the full twelve hours of exposure, the CO mass fraction at the miners' position is found to be 9.9062×10^{-4} . Since this is below 0.001 and the CO concentration never exceeds 0.02, the miners survive and escape the mine.

3 QUESTION 3: NUMERICAL SOLUTION EXTENDED

3.1 Source term in the numerical solution

MATLAB code implementing the source term $f(x, t) = -0.001 e^{-0.01 x^{1.5}}$ within the FTCS scheme from equation (32):

```

1 % Discretize the source term f(x,t) = -0.001e^(-0.01 x^(1.5))
2 f = -0.001 * exp(-0.01 * x.^1.5);
3
4 % Time-stepping loop
5 for t = dt:dt:t_final
6     % Initialize temporary vector for the new time step
7     c_new = c_numerical;
8
9     % Loop over space
10    for i = 2:Nx - 1
11        % Apply FTCS scheme with source term
12        c_new(i) = c_numerical(i) + sigma * (c_numerical(i - 1) ...
13            - 2 * c_numerical(i) + c_numerical(i + 1)) + dt * f(i);
14    end

```

The impact of the source term on the CO mass fraction is most significant near $x = 0$, due to the exponential decrease of the source term with distance (as a consequence of $e^{-0.01x^{1.5}}$). The impact diminishes as x increases.

3.2 Ventilated and Unventilated Mineshaft

The plot in Figure 6 shows a direct correlation between ventilation and safety in the mineshaft. While both cases begin with a CO mass fraction of 1 at $x = 0$, the CO mass fraction in the ventilated shaft drops almost instantly by approximately half, while the unventilated shaft maintains a smaller decline. Consequently, miners in the ventilated shaft are exposed to less CO over the time period, giving them a better chance of escaping. The MATLAB code to produce this plot can be found in the Appendix as 3.1.

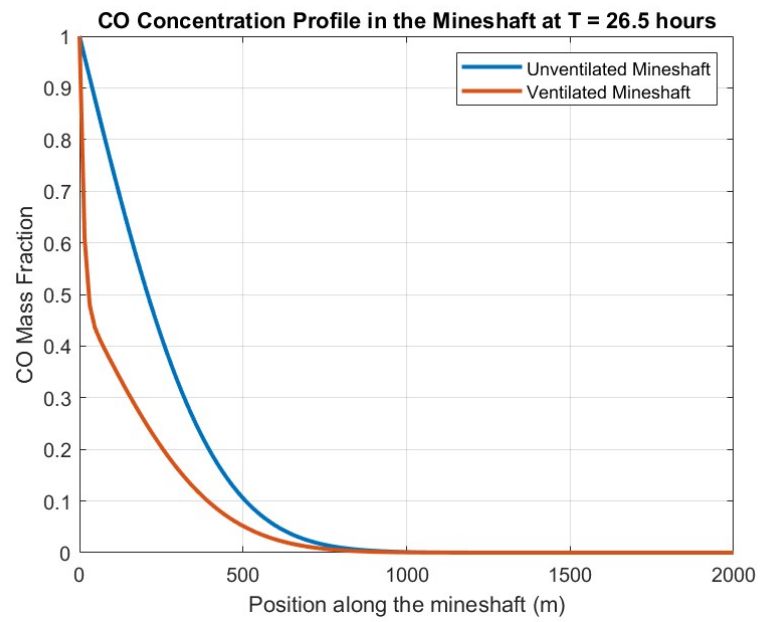


Figure 6: CO mass fraction in ventilated and unventilated mineshaft at $t = 26.5$ hr.

Appendix

1.7 Fourier Series Implementation

```

1 % Parameters
2 L = 2000; % Length of the mine shaft in meters
3 D = 5e-1; % Diffusivity of CO in m^2/s
4 t = 0; % t=0 as specified in Q1 .7
5 nfArray = [1 , 10 , 100]; % Three different nf values
6 % Spatial domain (129 Spatial grid points )
7 x = linspace (0 , L , 129) ;
8 % Modified initial condition
9 Y_initial = @( x ) x / L - 1;
10 % Plot the initial condition
11 figure ;
12 plot (x , Y_initial ( x ) , 'k', 'LineWidth', 2 , 'DisplayName', 'Mine
    shaft') ;
13 hold on ;
14 for nf = nfArray
15     % Fourier series approximation
16     Y_approx = zeros ( size ( x ) ) ;
17     for n = 1: nf
18         % Calculating Bn
19         Bn = -2 * mod(n,2) / ( n * pi ) ;
20         % Fourier series term contribution at tEnd
21         Y_approx = Y_approx + Bn * sin ( n * pi * x / L ) * exp ( - D *
            ( n * pi / L ) ^2 * t ) ;
22     end
23     plot (x , Y_approx , 'DisplayName', sprintf('Number of Fourier
        terms = %d', nf ) ) ;
24 end
25 % Customize the plot
26 title ( 'Fourier Series Approximation of the Modified Initial Condition
    ' ) ;
27 xlabel ( 'Position along the mine shaft , x (m)' ) ;
28 ylabel ( 'CO Mass Fraction , C_H (x ,0) ' ) ;
29 legend ( 'show', 'Location', 'north' ) ;

```

2.3 Comparison of Numerical and Analytical Solutions

```

1 clear all
2 clc
3 close all
4
5
6 nx = [9,17,33,65,129.];
7
8 for i = 1:length(nx)
9     x = linspace(0,2000,nx(i)); % x-domain [m]
10    Y = NumericalFTCS(nx(i));
11
12    Plot = figure(1);
13    plot(x,Y)
14    title ( 'CO Mass Fraction in the Mine Shaft at t = 30 min ' ) ;
15    xlabel ( 'Distance along the shaft (m)' ) ;
16    ylabel ( 'CO Mass Fraction ' ) ;
17    legend ( 'Nx = 9', 'Nx = 17 ', 'Nx = 33 ', 'Nx = 65 ', 'Nx = 129 ',
        ' Analytical Solution ', 'Location ', 'best ' ) ;

```

```

18
19     hold on
20 end

```

2.4 Error Norms and Order of Accuracy

```

1 clear all
2 clc
3 close all
4
5 %% Analytical and Numerical Soln
6 [C9A, dy9] = Analytical(9);
7 [C17A, dy17] = Analytical(17);
8 [C33A, dy33] = Analytical(33);
9 [C65A, dy65] = Analytical(65);
10 [C129A, dy129] = Analytical(129);
11
12 [C9N] = NumericalFTCS(9);
13 [C17N] = NumericalFTCS(17);
14 [C33N] = NumericalFTCS(33);
15 [C65N] = NumericalFTCS(65);
16 [C129N] = NumericalFTCS(129);
17
18 %% Error Norm
19
20 % nx = 9 row
21 L1_9 = norm(C9N - C9A, 1)*dy9;
22 L2_9 = norm(C9N - C9A, 2)*sqrt(dy9);
23 LInf_9 = norm(C9N - C9A, Inf);
24
25 % nx = 17 row
26 L1_17 = norm(C17N - C17A, 1)*dy17;
27 L2_17 = norm(C17N - C17A, 2)*sqrt(dy17);
28 LInf_17 = norm(C17N - C17A, Inf);
29
30 % nx = 33 row
31 L1_33 = norm(C33N - C33A, 1)*dy33;
32 L2_33 = norm(C33N - C33A, 2)*sqrt(dy33);
33 LInf_33 = norm(C33N - C33A, Inf);
34
35 % nx = 65 row
36 L1_65 = norm(C65N - C65A, 1)*dy65;
37 L2_65 = norm(C65N - C65A, 2)*sqrt(dy65);
38 LInf_65 = norm(C65N - C65A, Inf);
39
40 % nx = 129 row
41 L1_129 = norm(C129N - C129A, 1)*dy129;
42 L2_129 = norm(C129N - C129A, 2)*sqrt(dy129);
43 LInf_129 = norm(C129N - C129A, Inf);
44
45 %% Order of Accuracy
46
47 % n(L1) column
48 nL1_17 = (log10(L1_9/L1_17))/(log10(dy9/dy17));
49 nL1_33 = (log10(L1_17/L1_33))/(log10(dy17/dy33));
50 nL1_65 = (log10(L1_33/L1_65))/(log10(dy33/dy65));
51 nL1_129 = (log10(L1_65/L1_129))/(log10(dy65/dy129));
52

```

```

53 % n(L2) column
54 nL2_17 = (log10(L2_9/L2_17))/(log10(dy9/dy17));
55 nL2_33 = (log10(L2_17/L2_33))/(log10(dy17/dy33));
56 nL2_65 = (log10(L2_33/L2_65))/(log10(dy33/dy65));
57 nL2_129 = (log10(L2_65/L2_129))/(log10(dy65/dy129));
58
59 % n(LInf) column
60 nLInf_17 = (log10(LInf_9/LInf_17))/(log10(dy9/dy17));
61 nLInf_33 = (log10(LInf_17/LInf_33))/(log10(dy17/dy33));
62 nLInf_65 = (log10(LInf_33/LInf_65))/(log10(dy33/dy65));
63 nLInf_129 = (log10(LInf_65/LInf_129))/(log10(dy65/dy129));

```

2.5 CO Mass Fraction at Different Times

```

1 clear all
2 clc
3 close all
4
5
6 time = [1,6,12,18,24];
7
8 for i = 1:length(time)
9     x = linspace(0,2000,129); % x-domain [m]
10    Y = NumericalFTCS_time(time(i));
11
12    Plot = figure(1);
13    plot(x,Y)
14    title('CO Mass Fraction in the Mine Shaft at Different Times ');
15    xlabel('Distance along the shaft (m)');
16    ylabel('CO Mass Fraction , C');
17    legend('1 hours', '6 hours', '12 hours', '18 hours', '24 hours', '
Location ', 'best ');
18
19    hold on
20 end

```

2.6 Miner Survival Solver

```

1 C65Target = 0.001;
2 TimeVals = linspace(0, 48, 200); % Test times between 0 and 48 hours
3 PlaceholderTime = NaN;
4
5 for TimeInitGuess = TimeVals
6     [C, ~] = NumericalFTCS_time(TimeInitGuess);
7     % Check if concentration at C(65)/x=1000 reaches the target
8     if C(65) >= C65Target
9         PlaceholderTime = TimeInitGuess;
10        break; % Exit the loop if the target is reached
11    end
12 end
13
14 PlaceholderTime
15
16 [C,y] = NumericalFTCS_time(PlaceholderTime);
17 MinerPos750 = C(49);
18
19 [C,y] = NumericalFTCS_time(PlaceholderTime+(250/360));
20 MinerPos1000 = C(65);

```

```

21
22 [C,y] = NumericalFTCS_time(PlaceholderTime+(500/360));
23 MinerPos1250 = C(81);
24
25 [C,y] = NumericalFTCS_time(PlaceholderTime+(750/360));
26 MinerPos1500 = C(97);
27
28 [C,y] = NumericalFTCS_time(PlaceholderTime+(1000/360));
29 MinerPos1750 = C(113);
30
31 [C,y] = NumericalFTCS_time(PlaceholderTime+(1250/360));
32 MinerPos2000 = C(129);

```

Numerical FTCS Scheme

```

1 function [C,y] = NumericalFTCS(ny)
2 % Input :
3 %     ny - number of points in domain
4 % Output :
5 %     C - concentration of phosphate (mg/L) at end time
6 % Set constant paramters
7 L = 2000; % Length of domain [m]
8 D = 5E-1; % Diffusivity constant [m^2/s]
9 tf = 30*60;
10 % Generate range
11 y = linspace(0,L,ny); % y-domain [m]
12 dy = y(2) - y(1); % y-domain spacing [m]
13
14 MStepSize = ((L/256)^2)/(2*D)
15 % Calculate number of timesteps
16 nt = tf/MStepSize;
17
18 % Coefficient in FTCS scheme
19 sigma = D*MStepSize/dy^2;
20
21 % Initialise solution vector1
22 Cn = zeros(1,ny); % Solution at timestep n
23 Cnp1 = zeros(1,ny); % Solution at timestep n+1
24
25 % Set initial condition and boundary conditions
26 Cn(2:end) = 0;
27 Cn(1) = 1;
28 Cnp1(1) = 1;
29 Cnp1(end) = 0;
30
31 % Loop over time
32 for n = 2:nt
33     % Loop over internal points
34     for i = 2:ny-1
35         Cnp1(i) = Cn(i) + sigma*(Cn(i-1) - 2*Cn(i) + Cn(i+1));
36         Cnp1(ny) = Cn(ny) + sigma*(Cn(ny-1) - Cn(ny));
37     end
38     % Update solution for next timestep
39     Cn = Cnp1;
40 end
41
42 % Set output
43 C = Cnp1

```

44 **return**

Analytical Solution Function

```

1 function [C, y] = Analytical(ny)
2 % Input:
3 %   ny - number of points in domain
4 % Output:
5 %   C - concentration of phosphate (mg/L) at end time
6 %   y - spatial domain
7
8 % Set constant parameters
9 L = 2000;           % Length of domain [m]
10 D = 5E-1;          % Diffusivity constant [m^2/s]
11 tf = 30*60;        % End time [s]
12
13 % Generate spatial domain
14 y = linspace(0, L, ny); % y-domain [m]
15 dy = y(2) - y(1);      % y-domain spacing [m]
16
17 % Calculate analytical solution
18 C = zeros(1, ny);
19 for i = 1:ny
20     C(i) = 1 - erf(y(i) / (2 * sqrt(D * tf)));
21 end
22
23 end

```

Numerical FTCS for Distinct Time Periods

```

1 function [C,y] = NumericalFTCS_time(time)
2 % Input :
3 %   time - time of soln (in hours)
4 % Output :
5 %   C - concentration of phosphate (mg/L) at end time
6 % Set constant paramters
7 L = 2000;           % Length of domain [m]
8 D = 5E-1;          % Diffusivity constant [m^2/s]
9 ny = 129;
10 tf = time*60*60;
11 % Generate range
12 y = linspace(0,L,ny); % y-domain [m]
13 dy = y(2) - y(1);      % y-domain spacing [m]
14
15 MStepSize = ((L/256)^2)/(2*D);
16 % Calculate number of timesteps
17 nt = tf/MStepSize;
18
19 % Coefficient in FTCS scheme
20 sigma = D*MStepSize/dy^2;
21
22 % Initialise solution vector
23 Cn = zeros(1,ny); % Solution at timestep n
24 Cnp1 = zeros(1,ny); % Solution at timestep n+1
25
26 % Set initial condition and boundary conditions
27 Cn(2:end) = 0;
28 Cn(1) = 1;

```



```

29 Cnp1(1) = 1;
30 Cnp1(end) = 0;
31
32 % Loop over time
33 for n = 2:nt
34     % Loop over internal points
35     for i = 2:ny-1
36         Cnp1(i) = Cn(i) + sigma*(Cn(i-1) - 2*Cn(i) + Cn(i+1));
37         Cnp1(ny) = Cn(ny) + sigma*(Cn(ny-1) - Cn(ny));
38     end
39     % Update solution for next timestep
40     Cn = Cnp1;
41 end
42
43 % Set output
44 C = Cnp1;
45 return

```

3.1 Ventilated and Unventilated Mineshaft Comparison

```

1 function ThreeOne
2     % Define parameters
3     Nx = 129; % Number of spatial grid points
4     L = 2000; % Length of the mineshaft in meters
5     dx = L / (Nx - 1); % Spatial step size
6     D = 5e-1; % Diffusivity of CO in m^2/s
7     dt = 60; % Time step in seconds
8     t_final = 26.5 * 3600; % Final time in seconds (26.5 hours
    converted to seconds)
9
10    % Generate the spatial grid
11    x = linspace(0, L, Nx);
12
13    % Calculate numerical solution without source term for comparison
14    c_numerical = numericalSolutionFTCS(Nx, dt, dx, D, t_final);
15
16    % Calculate numerical solution with source term
17    c_numerical_ventilated = ventilated_numericalSolutionFTCS(Nx, dt,
    dx, D, t_final, x);
18
19    % Plotting
20    figure;
21    plot(x, c_numerical, 'LineWidth', 2, 'DisplayName', 'Unventilated
    Mineshaft');
22    hold on;
23    plot(x, c_numerical_ventilated, 'LineWidth', 2, 'DisplayName', '
    Ventilated Mineshaft');
24    hold off;
25    xlabel('Position along the mineshaft (m)');
26    ylabel('CO Mass Fraction');
27    title('CO Concentration Profile in the Mineshaft at T = 26.5 hours'
    );
28    legend('show');
29    grid on;
30 end
31
32 function c_numerical = ventilated_numericalSolutionFTCS(Nx, dt, dx, D,
    t_final, x)

```

```

33 % Initialize concentration profile
34 c_numerical = zeros(Nx, 1); % Solution vector
35 c_numerical(1) = 1; % Initial CO mass fraction fixed at 1.0 at x =
0
36
37 % Calculate coefficient for the FTCS scheme
38 sigma = D * dt / dx^2;
39
40 % Discretize the source term f(x,t) = -0.001e^(-0.01 x^(1.5))
41 f = -0.001 * exp(-0.01 * x.^1.5);
42
43 % Time-stepping loop
44 for t = dt:dt:t_final
45     % Initialize temporary vector for the new time step
46     c_new = c_numerical;
47
48     % Loop over space
49     for i = 2:Nx - 1
50         % Apply FTCS scheme with source term
51         c_new(i) = c_numerical(i) + sigma * (c_numerical(i - 1) - 2
* c_numerical(i) + c_numerical(i + 1)) + dt * f(i);
52     end
53
54     % Apply boundary conditions
55     c_new(1) = 1; % CO mass fraction fixed at 1.0 at x = 0
56     c_new(end) = 0; % CO mass fraction fixed at 0 at x = L
57
58     % Update concentration profile
59     c_numerical = c_new;
60 end
61 end
62
63 function c_numerical = numericalSolutionFTCS(Nx, dt, dx, D, t_final)
64 % Initialize concentration profile
65 c_numerical = zeros(Nx, 1); % Solution vector
66 c_numerical(1) = 1; % Initial CO mass fraction fixed at 1.0 at x =
0
67
68 % Calculate coefficient for the FTCS scheme
69 sigma = D * dt / dx^2;
70
71 % Time-stepping loop
72 for t = dt:dt:t_final
73     % Initialize temporary vector for the new time step
74     c_new = c_numerical;
75
76     % Loop over space
77     for i = 2:Nx - 1
78         % Apply FTCS scheme without source term
79         c_new(i) = c_numerical(i) + sigma * (c_numerical(i - 1) - 2
* c_numerical(i) + c_numerical(i + 1));
80     end
81
82     % Apply boundary conditions
83     c_new(1) = 1; % CO mass fraction fixed at 1.0 at x = 0
84     c_new(end) = 0; % CO mass fraction fixed at 0 at x = L
85
86     % Update concentration profile

```

```
87         c_numerical = c_new;  
88     end  
89 end
```